Enabling Flash and Boot of Qualcomm QTI PROD Secured Images on Old Hardware:

How to Fuse OEM_ID

Version History

Version	Date	Content
1.0	2022-05-30	Describe necessary workaround due to `CMCONTI-820`
1.1	2022-05-31	Correct dumpreg address – use address of sa515, not sa415
1.2	2022-05-31	Emphasize the need to use that “special” **`sec.elf`** image

Background

To fix the "trustzone image kernel panic" issue `CMCONTI-820`:

VW_CONMOD-1375 - [CMCONTI-820] [SWDL] SWDL fails due to kernel panic during activation stage VERIFYING

Qualcomm provided a new "`tz.mbn`" binary image as part of this fix, with increased size of heap memory. QC clamied they could only provide QTI PROD secured images = bound to an OEM_ID. No "unbound" images can be delivered any more.

This change was first introduced with SW **conmod-sa515m-3.3.138.0**.

As a consequence, any hardware this"`tz.mbn`" is to be flashed to must have the OEM_ID fuse burnt. Otherwise flashing will abort after the reboot to the newly flashed bank/images with the following trace fingerprint:

```
[5.25 08:34:52.7]D -       762 - Segments hash check
[5.25 08:34:52.7]D -     11133 - QSEE Dev Config Image Loaded, Delta - (37343
Bytes)
[5.25 08:34:52.7]B -    543662 - QSEE Image Load, Start from (tz)
[5.25 08:34:52.7]D -      5154 - Auth Metadata
[5.25 08:34:52.7]B -    556503 - Error code 100302e at boot_config.c Line 275
```

This will apply to "older" HW (e.g. B samples) that did not have this fuse burnt in production. With e.g. C2.1 there is no problem.

> **Note**: This fuse and secure image are NOT related to the "Secure Boot" feature and "prod-secured hardware". The problem that this "`tz.mbn`" cannot be flashed, applies to unsecured HW as well, if the OEM_ID fuse it not burnt.

There are two possible workarounds.

Workaround 1: Discontinue Use of B-Samples for Tests

Do not use "old" B samples for testing any longer, but use newer C-Hardware.

Workaround 2: Blow Fuse

On affected units, blow the OEM_ID fuse, setting it to 0x106. Follow these instructions:

1. Check the fuse value with command (register address for sa515 HW):

```
dumpreg 0x7801d0 16
```

2. If output is only 0:

```
007801c0: 00000000 00000000 00000000 00000000
```

3. Flash a special sec.elf using fastboot (origin of image see below):

```
sys_reboot booloader
fastboot flash sec sec.elf
```

4. Check the fuse value again. Output should be:

```
007801c0: 00000000 00000106 00000000 00000000
```

IMPORTANT:

- The special **`sec.elf`** binary image to be used is delivered as part of the SDK user documentation **`"CMCONTI-820_workaround_OEM_ID_fuse_sec_elf.zip"`** in subfolder:
  `\Documents\Attachments`
  (!) Do NOT use a sec.elf image from a generated 4K folder, since this would activate secure boot **(!)**
- For a more detailed instruction how to blow the OEM_ID fuse, please refer to the Appendix.

Appendix: Detail Description “Blow OEM Fuse”

How to blow OEM_ID on SA415 and SA515 chipsets?

- 1. OEM_ID overview
  - 1.1 What is the OEM_ID?
  - 1.2 Why is it needed?
  - 1.3 Download
  - 1.4 How does it look like?
- 2. Blow OEM_ID fuse

> **NOTE**
>
> Section 2 will show you how to blow the OEM_ID, but first download the right sec.file from section 1.3 Download

1. OEM_ID overview

OEM_ID must be flashed on all the existing SA415 and SA515 products. This procedure is a one time flash and once the fuses are blown, they'll keep the same value for all the subsequent flash procedures.

1.1 What is the OEM_ID?

OEM_ID is an identifier that Qualcomm has assigned Continental Automotive among all its customers.

The assigned value to Continental Automotive is: **0x0106** and this is a one time flash. Once the fuses are blown, they'll remain like this for ever.

1.2 Why is it needed?

Qualcomm will provide production certificates to enable what they refer to as *premium features* for each platform (e.g. SA415 and SA515, etc.). Examples include features like:

1. GPS L1
2. GPS L1+L5
3. CV2X
4. PPE vendors (e.g. Trimble RTx)

1.3 Download

> **IMPORTANT**
>
> **Be aware:** Select the proper sec.elf file. Flashing SEC file has no rollback and **changes are permanent**.

	SA415m	SA515m
OEM_ID sec file	sec.elf	sec.elf

1.4 How does it look like?

The attached sec.elf files contains the following fuse entries to blow:

```
/* SA415m OEM_ID address */
Fuse Entries:
| S.No | Address | Operation |   MSB    |   LSB    | Region Type |
|------|---------|-----------|----------|----------|-------------|
|  1   | 00780190|    BLOW   | 00000106 | 00000000 |  OEM_CONFIG |

/* SA515m OEM_ID address */
Fuse Entries:
| S.No | Address | Operation |   MSB    |   LSB    | Region Type |
|------|---------|-----------|----------|----------|-------------|
|  1   | 007801d0|    BLOW   | 00000106 | 00000000 |  OEM_CONFIG |
```

OEM_ID gets printed as part of the bootloader logs (see line 11).

```
OEM_ID gets printed as part of the early boot up logs in the A7 core (SBL logs):
Format: Log Type - Time(microsec) - Message - Optional Info
Log Type: B - Since Boot(Power On Reset),  D - Delta,  S - Statistic
S - QC_IMAGE_VERSION_STRING=BOOT.SBL.4.0-00119
S - IMAGE_VARIANT_STRING=MAATANAZA
S - OEM_IMAGE_VERSION_STRING=telematics-int16
S - Boot Interface: NAND
S - Secure Boot: Off
S - Boot Config @ 0x00786070 = 0x00000021
S - JTAG ID @ 0x00786130 = 0x0011d0e1
S - OEM ID @ 0x00786138 = 0x00000000
S - Serial Number @ 0x00786134 = 0x2345fcf3
S - OEM Config Row 0 @ 0x00784188 = 0x0000000000000000
S - OEM Config Row 1 @ 0x00784190 = 0x0000000000000000
S - Feature Config Row 0 @ 0x007841a0 = 0x0040000000280b80
S - Feature Config Row 1 @ 0x007841a8 = 0x0000000002800000
S - Continental HWID @ 0x00784590 = 0x0000000000080200
S - QFuse Write Permission LSB @ 0x00784158 = 0x02a4e804
```

On the logs from above (on line 11), OEM ID is set to zeroes.

Once you blow the fuses (by following Section 2), you'll see a difference on the value for this identifier.

```
Format: Log Type - Time(microsec) - Message - Optional Info
Log Type: B - Since Boot(Power On Reset),  D - Delta,  S - Statistic
S - QC_IMAGE_VERSION_STRING=BOOT.SBL.4.0-00119
S - IMAGE_VARIANT_STRING=MAATANAZA
S - OEM_IMAGE_VERSION_STRING=telematics-int16
S - Boot Interface: NAND
S - Secure Boot: Off
S - Boot Config @ 0x00786070 = 0x00000021
S - JTAG ID @ 0x00786130 = 0x0011d0e1
S - OEM ID @ 0x00786138 = 0x01060000
S - Serial Number @ 0x00786134 = 0x2345fcf3
S - OEM Config Row 0 @ 0x00784188 = 0x0000000000000000
S - OEM Config Row 1 @ 0x00784190 = 0x0000010600000000
S - Feature Config Row 0 @ 0x007841a0 = 0x0040000000280b80
S - Feature Config Row 1 @ 0x007841a8 = 0x0000000002800000
S - Continental HWID @ 0x00784590 = 0x0000000000080200
S - QFuse Write Permission LSB @ 0x00784158 = 0x02a4e804
```

OEM_HW_ID is part of the MSB section, in the other hand OEM_PRODUCT_ID is part of the LSB section.

You can also check the fuse value using **dumpreg** application once A7 gets booted up:

```
[root@sa415m ]#dumpreg -h
dumpreg -h
dumpreg <addr> <len
[root@sa415m ]#dumpreg 0x780190 16
dumpreg 0x780190 16
00780180:                                    00000000 00000000 00000000 00000000
[root@sa415m ]#

[root@sa415m ]#

[root@sa415m ]#sys_reboot bootloader
sys_reboot bootloader

C:\Users\uidj3209>cd Documents\nad_shared

C:\Users\uidj3209\Documents\nad_shared>fastboot flash sec sec.elf
writing 'sec'...
OKAY [  0.003s]
sending 'sec' (12 KB)...
OKAY [  0.034s]
finished. total time: 0.041s

C:\Users\uidj3209\Documents\nad_shared>adb shell

#################################################################
# Copyright (C) 2020 Continental Automotive Systems, Inc.
#
# Chestnut Development Board
#
# Platform Version: OTP-SA415M-3.1.6.11-DEVEL
# Linux Version:    4.9.213LE.UM.3.2.1-03010-sa415m-100-g7e2ace5
# NAND info:        2680AC2C 512M/256K/4K/8b
# RAM info:         299248K /357620K available
# Build Date:       Wed Jun  3 03:57:08 UTC 2020
#
# WARNING! Development: Console always allowed
#           (DO NOT PROVIDE IMAGE TO OEM!)
#################################################################

[root@sa415m ]#dumpreg 0x780190 16
dumpreg 0x780190 16
00780180:                                    00000000 00000106 00000000 00000000
[root@sa415m ]#
```

2. Blow OEM_ID fuse

In order to blow the OEM_ID fuse, you must boot in fastboot mode. To enter this mode you have to:

1. Open a CMD window and send: **adb reboot bootloader**
2. Under the same CMD window, wait for the board for boot up and send: **fastboot devices**
   a. You'll see something like this:

```
C:\Users\uidj3209\Documents\nad_shared>fastboot devices
2345fcf3        fastboot
```

3. Your device is ready to receive fastboot commands. In order to flash the new SEC image, send the following command: **fastboot flash sec sec.elf**
   a. Make sure you are sitting on the sec.elf folder.

```
C:\Users\uidj3209\Documents\nad_shared>fastboot flash sec sec.elf
writing 'sec'...
OKAY [  0.003s]
sending 'sec' (12 KB)...
OKAY [  0.034s]
finished. total time: 0.041s
```

4. Once the finished log gets printed, you can boot into normal mode again. Send: **fastboot reboot**